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RESEARCH ARTICLE

Nano-hydroxyapatite/β-tricalcium phosphate ceramics scaffolds loaded with cationic liposomal ceftazidime: preparation, release characteristics *in vitro* and inhibition to *Staphylococcus aureus* biofilms

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Abstract

In the current study, nano-hydroxyapatite/β-tricalcium phosphate (HA/β-TCP) ceramics scaffolds loaded with cationic liposomal ceftazidime (CLCs) prepared by modified reverse phase evaporation method, the investigations of their release characteristics were performed by the dissolution tests, *in vitro* anti-biofilm activity of the scaffolds was studied by the determination of bacterial susceptibility with ELISA. The mean particle size, zeta potential, pH and entrapment efficiency of the CLCs studied were 161.5 ± 5.37 nm, 60.60 ± 5.24 mV, 6.90 ± 0.07 and 16.57 ± 0.13%, respectively. Electron microscopic images of the samples indicated that the liposomes were well preserved in the scaffolds and that it was the CLCs rather than free ceftazidime releasing from the scaffolds. The minimal inhibitory concentrations (MICs) to *Staphylococcus aureus* of free ceftazidime and its liposomal formulation were 6.00 μg/mL and the release behaviors of both CLCs and free ceftazidime from scaffolds were based on the dissolution/diffusion processes, Fick's law. These results demonstrated that CLCs could inhibit remarkably the formation of *S. aureus* biofilm more effectively than free ceftazidime ($P < 0.05$). The study demonstrated that the HA/β-TCP ceramic scaffolds was such a material that could sustain release CLCs and maintain the adequate amounts of CLCs to absorb to biofilm. It provided an ideal way to inhibit bacterial biofilms for clinical practices.

Keywords: Bone regeneration, HA/β-TCP ceramic scaffold, *Staphylococcus aureus* biofilm, liposomal ceftazidime, controlled drug release

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Background

Osteomyelitis is a progressive infection of bone that results in inflammatory destruction of the bone, bone necrosis etc. and may lead to a chronic and persistent state[1]. *Staphylococcus aureus* is the most common organism isolated from patients with bacterial osteomyelitis[2]. *S. aureus* will evolve antibiotic resistance rapidly with forming biofilms[1].

Commonly, biofilms of bacteria do not resist directly to antimicrobials but tolerate them, bacteria become sensitive to antimicrobials when returning to the planktonic state[3]. The insensitivity of bacteria biofilm to antibiotics depend on their cell wall composition, surface structures, and phenotypic variation in enzymatic activity[4,5]. The main composition of biofilm is glycocalyx which is typically anionic in nature[6], and this has been suggested that the negatively charged exopolysaccharide is very effective in protecting bacterial cells from cationic antibiotics by inhibiting their permeation[7].

From previous studies, it has been confirmed that antibiotic-loaded liposomes could against colonising microorganisms[2,8], enrich antibiotics at biofilm interfaces[9,10].

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Cationic liposomes were prepared by phosphatidylcholine, cholesterol and polyammonium derivatives, such as stearamide and dipalmitoylphosphatidylethanolamine[11–13]. The liposomal surface become positive-charged with modification of the ammonium derived polymers. The cationic liposomes could incorporate adequate quantity of antimicrobial agents[14]. It is reasonable to speculate that the cationic liposomes loaded with antimicrobiotics provide a new way to treat cryptic bacterial infection[15].

For the treatment of osteomyelitis, local antibiotic therapy has several advantages over the systemic administration of antibiotics[2]. Systemic intravenous administration of antibiotics would necessarily take 4–6 weeks for antimicrobial therapy. It would increase considerably inherent risks of severe systemic complications[16]. It was reported that the earliest local treatment of osteomyelitis was gentamicin beads, but it need additional operations to remove the beads within 2–4 weeks[17]. The ideal bone filler material should be biodegradable and adequate local antibiotics, at the same time boost bone regeneration[18]. Hydroxyapatite (HA) can promote bone growth around the implant and play an important role in the overall physiological functions and in the osseointegration process[19]. Because of its poor mechanical properties, the pure HA must be combined with other materials for the orthopedics in clinical applications, such as β-tricalcium phosphate (β-TCP), which possesses adequate strength for bone reconstruction and has proven to be osteoinductive and osteoconductive[20,21]. Combined applications of HA and β-TCP could form scaffolds to regenerate more effectively the inflammatory injure of the bone caused by osteomyelitis.

The objective of this study is to prepare the clinical applicable HA/β-TCP ceramic scaffolds loaded with liposomal ceftazidime, to investigate their release behaviors in the simulated body fluid (SBF)[22] and their anti-biofilms properties *in vitro*.

Methods

Materials and equipments

Soya phosphatidylcholine (SPC, Lot No. 090308) was obtained from Shanghai Huishi Biochemical Reagent Co. (China). Cholesterol (Chol, Lot No. 75U10120, ≥99.5%), Triton X-100 (Lot No. 87700150, >99%), phosphate buffered saline (PBS, pH = 7.40, Lot No. 95E00110) were purchased from Beijing Dingguo Biological Technology Co. (China). Stearamide (SA, Lot & filling Code 1322932 43907148) was purchased from Sigma (USA). Phosphatidyl glycerol (PG, L-α-Phosphatidyl-DL-glycerol, Lot 20090310) was purchased from Avanti Polar Lipids Inc. (USA). Sephadex G-50 (dry bead diameter 50–150 μm, bed volume 9–11 mL/g, Lot No. 080128) was bought from Shanghai Yuanju Bio-Tech Co. (China). Control ceftazidime (84.8%. Lot 130484-200803) was bought from National Institute of the Control of Pharmaceutical and Biological Products (China). Ceftazidime injection (Lot No. 080101) was provided by Kunming Ji Da Pharmaceutical Co., Ltd. (China). Bacterial strain of *S. aureus* was obtained from the Laboratory of General Hospital of Kunming Military Region. Brain heart infusion (BHI, Lot No. 200904231) was from Guangdong Huankai Microbial Science and Technology Co., Ltd.(China). Bacteriological agar (Lot No. 090406) was purchased from Hangzhou Tianhe Microorganism reagent Co., Ltd.(China). Yeast extract powder (Lot No. 20090813) was purchased from Oxoib Ltd. (UK). Water was distilled doubly. Reagents used for high performance liquid chromatography (HPLC) were chromatography pure grade (Sigma, USA). Microporous HA/β-TCP scaffolds(medical grade) were amicably donated by College of Materials Engineering, Kunming University of Science and Technology. All other chemicals were of analytical grade. HPLC system: Waters Aliliance 2695 with Waters 2489 and Empower software(Waters, USA), Waters symmetry C_{18} Column (4.6 mm × 250 mm, 5 μm, Waters, USA).

Scaffolds preparation

Preparation and characteristics of cationic liposomal ceftazidime (CLCs)

CLCs were prepared by modified reverse phase evaporation method[23]. The lipid mixture used for preparing liposomes was composed of SPC, SA and Chol at the optimized molar ratios of 7:1:1 and dissolved in mixed solvent of chloroform–ethanol (99:1) in a pear-shaped flask at 35°C. The ceftazidime buffer solution was directly added to organic phase (organic phase:aqueous phase 3:1, v/v), and sonicated for 10 min at 0°C in an ultrasonic bath (AS20500A, AUTO Science, Tianjin, China) to form an emulsion, followed by evaporation of organic solvent by a rotary evaporator (R-210, BUCHI, Switzerland) under vacuum at 0°C. Finally the liposomal suspension was filtered through 0.2 μm polycarbonate filter membrane three times. All products were quickly sealed in glass containers and stored in the darkness at 4°C.

The liposomes prepared above were negatively-dyed with 1.0% phosphotungstic acid and their microstructure was observed by transmission electron microscope (JEM-1010, Japan) (Figure 1).

The liposomes were diluted 10-fold with PBS. Determination of their particle sizes and ξ potentials of each liposome were performed three times at room temperature with Zetasizer (Zetasizer-Nano ZS, Malvern, UK).

The separation of liposomes were performed via mini-column centrifugation by Sephadex G-50 as described by Fry[24]. The samples were placed in the mini-column of Sephadex G-50, excessive PBS was first removed from the Sephadex column by centrifugation at 3000 rmp for 3 min. 0.25 mL of the liposome suspension was added to the mini-column and centrifuged with the same manner. The centrifugation solution was collected.

The lipid membranes of the liposomes were destroyed using 0.2% (w/v) Triton X-100 and the concentrations of the antibiotic in the liposomes were determined by HPLC at 254 nm with the mobile phase of triethylamine-glacial

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acetic acid buffer solution (pH5.0)-acetonitrile (93:7, v/v) that was adjusted to a pH 5.0 with acetic acid, and the flow rate of 1.0 mL/min under 35°C. The testing volume was 10 μL.

The entrapment efficiency (EE) of the antibiotic was calculated as follow:

$$EE(\%) = \frac{\text{Ceftazidime in liposomes}}{\text{Total ceftazidime in solution}} \times 100\% \quad (1)$$

Preparation of HA/β-TCP ceramic scaffolds loaded by CLCs

Microporous HA/β-TCP scaffolds were immersed in CLCs suspension and evaporated by a rotary evaporator to allow CLCs into the porous scaffolds under vacuum (0.09 MPa) at 0°C for about 1 h, followed by transfering into glass bottle and lyophilization.

For determination of CLCs' contents in scaffolds, the granules were ground in 1% Triton X-100 solution and then sonicated for 10 min to let ceftazidime completely release from the CLCs. The resulting suspension was centrifugated at 4000 rpm for 10 min, and the ceftazidime content in the supernatant was determined by HPLC as described above.

The morphologies of CLCs-loaded scaffolds were observed using a scanning electron microscope (S-3000N, HITACHI, Japan).

In vitro release characteristics of ceftazidime of the scaffolds

The *in vitro* release characteristics of ceftazidime of the scaffolds were determined with a dissolution apparatus(ZRS-6G Intelligent Dissolution Tester, Tianda Tianfa Technology Co. Ltd., China). A scaffold was soaked into 50 mL SBF solution at 37.0°C and stirred at 50 rpm. 0.5 mL of solution was withdrawn at fixed time interval. The ceftazidime concentrations of samples were analyzed three times by HPLC method described above.

In order to validate the release form as liposome from scaffolds, we impregnated the scaffolds in 10 mL PBS for 1 h, and observed 0.5 mL SBF solution by TEM.

Bacterial cultures and of biofilms preparations

Preparation of agar plates[25]: BHI was completely dissolved in water. Bacterial agar was then added to BHI solution and dissolved fully by boiling. The broth was autoclaved at 121°C for 15 min. After cooling to 40–45°C, the agar solution was poured into sterile petri dishes and solidified at 37°C.

Preparation of nutrient broth: 0.3% w/v yeast extra powder was added into BHI, completely mixed and then autoclaved.

Bacterial strain of *S. aureus* were cultured on the agar plates by streaking the organism to form single colony. The plates were inverted and incubated overnight at 37°C. Single colony of the organism was inoculated in aliquots (30 mL) of broth. The broth was incubated at 37°C for 18 h. The bacterial suspension was centrifuged at 60,000*g* and the supernatant was removed and replaced with PBS. The pellet was re-suspended and centrifuged twice. The bacterial concentration was adjusted by diluting with PBS to give an absorbance of 0.5 at 550 nm. This cell dispersion was used to prepare immobilised biofilm on the wells of the 96-well flat-bottomed microtiter plate. Two hundred micro liters of bacterial suspension was planted into a 96-well flat-bottomed microtiter plate and incubated at room temperature to form adsorbed biofilms of close-packed cells. After adsorption, the bacterial suspension was poured off, and the adsorbed biofilms were washed three times with sterile PBS.

Determination of bacterial susceptibility

The antimicrobial activity of the liposomes was determined in comparison to the free drug, by determining the MIC according to ATCC standard strains (National Committee for Clinical Laboratory Standards, 2009). The MIC was determined by a broth microdilution assay. The nutrient broth (100 μL) was inoculated into a 96-well flat-bottomed microtiter plate. The stock solution of the liposome suspension (100 μL) was added to the first wells and serial 2-fold dilutions were performed with a multichannel pipette, beginning with the second well, and discarding the final 100 μL. Samples with 10 concentrations were prepared. A suspension of organisms (1 μL) was added to each well containing liposomes and control well. Plates were sealed and incubated at 37°C under atmospheric conditions up to 18 h. The MIC was defined as the lowest concentration of antibiotic inhibiting visible growth after 18 h. The tests were performed three times for every specimen.

Anti-biofilms activity

The biofilm susceptibility to samples was assessed using a regrowth activity assay[25]. The biofilms were prepared in a 96-well flat-bottomed microtiter plate according to the method described above. After washing with PBS, the scaffold or liposome suspension and 500 μL of PBS were added to each well. After incubation for set intervals at room temperature, the specimen and PBS were poured out. Then, the wells were washed three times with sterile PBS and 200 μL of sterile growth medium was added. The plate was sealed and incubated at 37°C for 18 h. The absorbance of the growth medium was measured at 630 nm by Rayto Automatic ELISA Analyzer (RT-6000, Rayto Life and Analytical Science Co., Ltd., China). The absorbances of the wells containing only bacteria and no bacteria (after 18 h exposure to growth medium) were set as 100% and 0% growth, respectively. Growth inhibition by the liposome suspension or the free drug was calculated according to the following equation:

$$Inhibitin(\%) = \frac{OD_{100} - OD_s}{OD_{100} - OD_0} \times 100\% \quad (2)$$

where OD_{100} represents the optical density value of 100% growth, OD_0 is the optical density value of 0% growth, and OD_s is the optical density of the specimen.

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The inhibition values of specimen were calculated by Eq. (2). The test was performed three times for each specimen.

Statistical analyses

The test data were expressed as mean ± SD. Statistical analyses were performed by Student's *t*-test. Results were considered significantly different if the $P < 0.05$.

Results

Characterization of CLCs

The mean particle size, zeta potential, pH value and antibiotic EE of processed CLCs were 161.5 ± 5.37 nm, 60.60 ± 5.24 mV, 6.90 ± 0.07 and 16.57 ± 0.13%, respectively. After resolved with PBS solution, the particle size of freeze-dried CLCs was 103.0 ± 2.62 nm, it was decreased compared with the unfreeze-dried CLCs suspension, because that the inner aqueous phase of CLCs had a decrease when freeze-dried.

Figure 1 shows the microstructures of CLCs and scaffolds as observed with electron microscope. The multilamellar lipid vesicle was the main form of CLCs observed by transmission electron microscope (JEM-1010, Japan) (Figure 1A-1 and A-2). Before having been loaded with CLCs, the scaffolds were rich in cracks and pores observed by scanning electron microscope (S-3000N, HITACHI, Japan) (Figure 1B-1 and B-2). After having been loaded with CLCs, the scaffolds changed in microstructures and their cracks and pores were filled with CLCs (Figure 1C-1). The microstructures of CLCs could be clearly seen in magnified images of scaffolds (Figure 1C-2), demonstrating that CLCs had been well preserved in processed scaffolds.

Compared the CLCs released from scaffolds (Figure 1D-1) with freeze thawing CLCs (Figure 1D-2), they had almost same shape, and their particle size was smaller than normal CLCs, these showed that the materials released from scaffold were mainly in the form of CLCs instead of free ceftazidime.

In vitro release of CLC out of scaffolds

Due to the fact that the modified scaffolds have a heterogeneous and complex structures, the release mechanism should be complex. Frutos et al.[26,27] concluded that release of free antibiotics from scaffolds was based on the dissolution/diffusion processes and included three stages: (1) The antibiotic particles attached to the external surface of the scaffold were washed off or released at a large burst in a short time. (2) The solvent caused the fracture of external surface of the scaffold, resulting in gaps or orifices which allowed the solvent to penetrate into the scaffold and the antibiotic incorporated in the scaffold to release. This was a diffusion-controlled process. (3) Drug molecules impregnated in the scaffold were dissolved and released through capillaries or channels in the scaffold. They proposed a semi-empirical equation as following:

$$\frac{M_t}{M_\infty} = c + at^{\frac{1}{2}} + b(1-e^{-nt}) = c + at^{1/2} + b\left(1-e^{-nt}\right) \qquad (3)$$

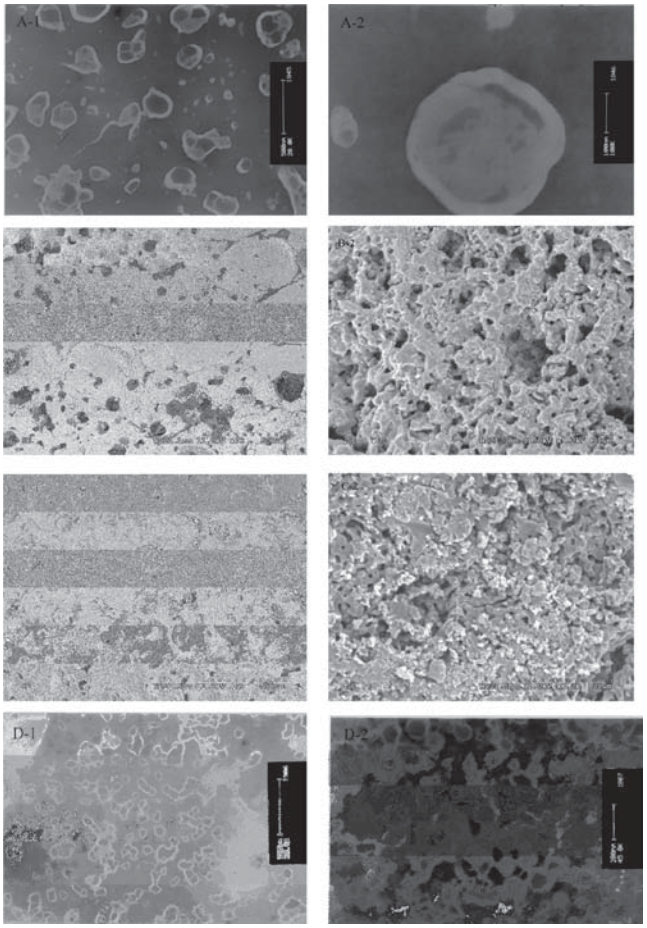


Figure 1. Electron micrographs (A) CLCs (B) Free scaffold (C) CLCs-loaded scaffold; scaffolds were impregnated in SBF solution for 1 h and then 0.5 mL SBF solution were sampled to observe (D-1); freeze thawing CLCs (D-2).

where $\frac{M_t}{M_\infty}$ is the cumulative ratio of antibiotic released from scaffold at time *t*, *c* is associated with the initial surface wash-off or burst effect, *1/2* represented a Fick's diffusion process, and the third process described above is represented by $b(1-e^{-nt})$. The parameters *a* and *b* represent the contributions (%) of the dissolution and diffusion processes to the release of liposomes or free drug at time *t* [26].

Our research demonstrated that the drug was released in the same manner. The parameters obtained for this model and their correlation coefficients were shown in Table 1. The correlation coefficient (r^2) implied that the release profiles of scaffolds (Figure 3) fitted well to Eq. (3). A large initial burst release of antibiotic was observed in the initial several hours of the experiments, and a high dose of the liposomal antibiotic released in a short time is beneficial to intercept the development of bacterial biofilm in early stage, which can largely reduce the risk of refractory bone infections such as osteomyelitis. From Figure 3, we could observe that nearly 100% antibiotic released from scaffolds within 72 h. For this reason the risk of bacterial resistance should be reduced. Other

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Table 1. Statistical parameters and correlation coefficient r^2 according to Eq. (2) for the CLCs and free ceftazidime release data from HA/β-TCP ceramics scaffolds.

	Parameters of Eq. (1)				
Samples	c	a	b	n	Correlation coefficient r^2
CLCs-loaded scaffolds	0.1367	0.1367	0.6349	0.2674	0.9791
Free drug loaded scaffolds	0.2574	0.2574	0.4423	0.1429	0.9816

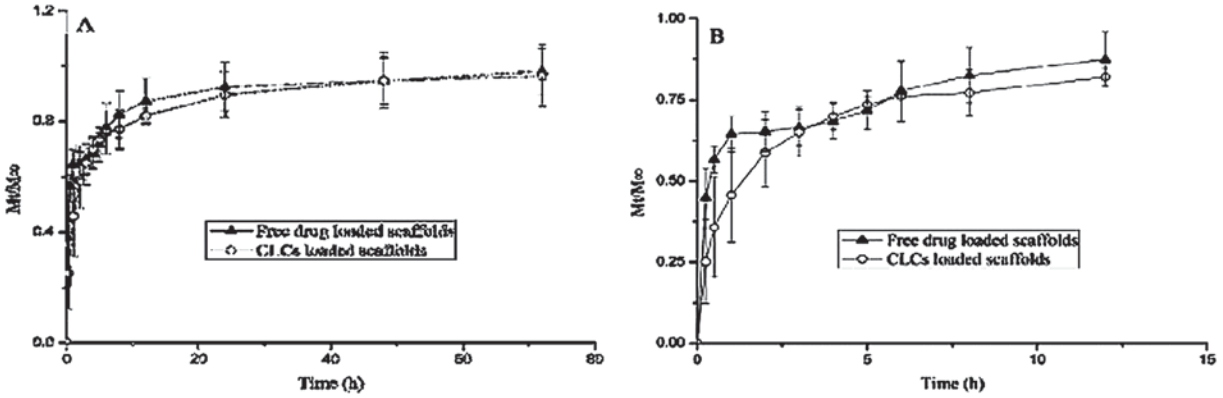


Figure 2. The release profiles of CLCs-loaded scaffolds and free drug loaded scaffolds (A). A large initial burst release of antibiotic was observed in all scaffold specimens (B).

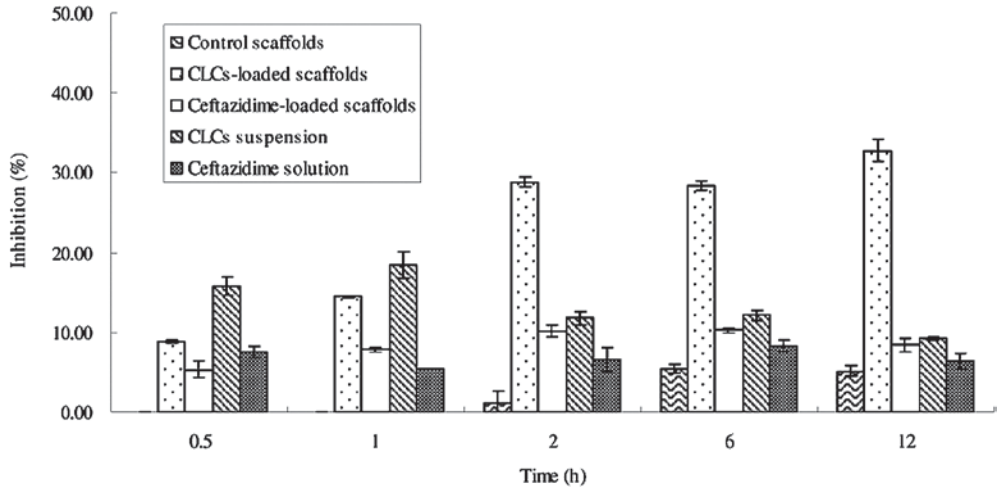


Figure 3. The results obtained for regrowth inhibition by CLCs-loaded scaffolds relative to free drug loaded scaffolds as a function of liposomal and free ceftazidime-biofilm incubation.

components degraded from scaffolds would be helpful to reconditioning the inflammatory destruction of the bone caused by bone infection.

Bacterial susceptibility

The MICs obtained from free drug and liposome formulations were 6.00 μg/mL and 4.25 μg/mL respectively. All test specimens were sensitive to the investigated strains from the tests. Therefore, liposome could be a suitable device for antimicrobial agents.

Anti-biofilms activity

The drug contents in control scaffolds, CLCs-loaded scaffolds, free drug loaded scaffolds, CLCs suspension and ceftazidime solution were 0, 1347.427, 1341.731, 1300.982 and 1291.682 μg respectively. Figure 4 showed the growth inhibitions by different samples. The inhibition of free scaffolds was below 6%, the free scaffolds showed some degree of absorption in biofilms. The results demonstrated that CLCs could inhibit significantly the formation of *S. aureus* biofilm more effectively than free drug ($P < 0.05$). The effectiveness of CLCs suspension increased with time of incubation, then became less after 60 min, because some liposomes may disrupt with increasing incubation time. The less retention of free antibiotic to the biofilms resulted in reducing inhibition. This result was similar to that as reported by Jones et al.[25]. These implied that only liposome suspension applied for inhibition of biofilm is limited. But for the CLC-loaded

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scaffolds, the release of CLCs from scaffolds was sustained, and these could provide the adequate CLCs to contact with biofilm.

Conclusions

Our study demonstrated that liposomal ceftazidime could be successfully incorporated into porous HA/β-TCP ceramics scaffolds. It was confirmed by electron microscope that liposomal ceftazidime instead of free ceftazidime was released from the scaffolds. Both CLCs and free ceftazidime released from scaffolds were based on the dissolution/diffusion processes, Flick's law. A large initial burst release of CLCs from scaffolds occurred to all scaffolds processed at the first 1-h of tests. The test results of all processed scaffolds fitted well to the Eq. (3). Release of nearly 100% antibiotic d from scaffolds within 72 h could reduce the risk of bacterial resistance. Other components degraded from scaffolds would be beneficial to repair of the inflammatory damage of the bone caused by bone infection.

Our study also confirmed that both ceftazidime and CLCs were susceptible to *S. aureus*. Moreover, CLCs could inhibit significantly the formation of *S. aureus* biofilm more effectively than free drug ($P < 0.05$). The effectiveness of CLC increased with time of incubation, but after 60 min the liposomal antibiotic became less effective, possibly due to that some liposomes may disrupt and part of ceftazidime may be released from liposomes. The free drug showed relatively lower efficacy for inhibiting biofilms at the same time. These imply that only application of liposome suspension for inhibition of biofilms is limited. In summary, the HA/β-TCP ceramics scaffolds could keep sustained release CLCs and these could provide the adequate amounts of CLCs to contact with the biofilms. In summary, from our studies, nano-hydroxyapatite/β-tricalcium phosphate ceramics scaffolds loaded with cationic liposomal ceftazidime provided an ideal strategy to locally inhibit bacterial biofilms for clinical practice.

Declaration of interest

The authors wish to record their gratitude to "11th Five-Year-Plan" Science and Technology Foundation of PLA (08G035) for financial support. T.H.Z. carried out the literature survey, designed and executed the project. M.S. and B.C.S. were involved in the interpretation of the data and writing the manuscript. T.M. and H.L.L. performed some of the experiments. Q.H.C. kindly provided the HA/β-TCP scaffolds. W.S. helped in sequence alignment and participated in proof-reading. Y.Q.X. and G.L.X. provided overall supervision and guidance for the project. All experiments were carried out at Kunming Medical College and Yunnan University. All authors read and approved the final manuscript.

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